

Ali Shariatmadari

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Experience

Robinhood – Software Engineer

Toronto, ON | May 2025 – Present

- Spearheaded Robinhood's migration to **zero-code** coin launches, built tooling to aggregate, enrich, and migrate **130+ assets** across multiple chains and jurisdictions into a single source of truth, eliminating **5h/week** in engineering work
- Re-architected the core crypto custody service from **Python** to **Go**, owning and productionized the **multi-tenant database routing** (data isolation across **5+** custodial entities with zero cross-tenant bleed), unified balance tracking across multiple custodial partners (**Fireblocks**, **BitGo**, **Coinbase Prime**), and core **AML** flow design
- Own Robinhood's cross-venue onchain crypto settlement stack end-to-end as the primary engineering POC for ops teams; expanded **WebAuthn**-attested settlement support to Europe, sped up the settlement calculation algorithm by **20×**, eliminated timeout-prone runs, and enabled batched settlements saving **25h/week** in manual work
- Built and launched crypto **cost-basis** (tax-lot), a product allowing users to track their tax lots, designed the **gRPC** API surface, validation, and a batched service layer that serves **800 QPS**; **30k+** lots created on launch week
- Helped bring Robinhood to **Canada** by delivering the crypto-custody and regulatory books-and-records integrations behind the launch, onboarding a new custodian and acquired Canadian platforms into the custody stack
- Held **monthly 24/7 on-call** shifts for crypto money-movement; triaged production incidents (notably **transaction-signing** failures), responded to cross-team requests, handled deployment issues and vendor outages, led multiple **SEVs** end-to-end, drove corrective actions, and presented SEV reviews to the entire crypto org; recovered **\$500,000** in customer funds

Bitgo – Software Engineering Intern

Palo Alto, CA | May 2024 – Sept 2024

- Implemented instant ACH deposit/withdrawals for retail customers in a purely functional code-base with **fp-ts**
- Built notification system for off-exchange asset settlements between partners using **TypeScript** and **fp-ts**

PostGrid – Software Engineering Intern

Toronto, ON | Jan 2024 – Apr 2024

- Created a tailored tracking feature for physical mail to increase customer conversions for clients using **React** and **Typescript**
- Sped up build and test time in CI by **2x** by implementing incremental caching strategies using **Github Actions** and **Docker**

Capsule – Software Engineering Intern

New York, NY | Sep 2022 – Dec 2022

- Built an EVM-powered web3 application for a virtual on-chain wallet using **Solidity**, **Hardhat** and **React**, deployed on IPFS

Bloq – Software Engineering Intern

Chicago, IL | Jan 2022 – Apr 2022

- Built Blockchain node/staking infrastructure on **AWS/Kubernetes**, JRPC/IPC proxy services, and **Node.js/Docker** tooling; expanded profiling coverage **30%** and improved backup generation speed **200%**
- Rearchitected the blockchain node API service to support asynchronous request processing and dynamic service routing using an **NGINX** ingress controller on a **Kubernetes** cluster, enabling rolling updates and improving scalability

TD Bank – Software Engineering Intern

Toronto, ON | May 2021 – Aug 2021

- Built a predictive banking dashboard and financial-data ingestion/visualization pipeline using **React**

Education

University of Waterloo – Bachelor of Software Engineering (BSE)

April 2025

- Courses: Machine Learning, Distributed Systems, Concurrency, Compilers, Graphics, Operating Systems, Algorithms

Projects

Counter-Strike 🚩 – 3D Game Engine

C++, OpenGL

- Built a real-time FPS in C++/OpenGL with dynamic lighting, physics, bullet impacts, texture mapping, models, and animations

CtrlAirSpace 🚩 – Air Mouse | Hack the North 2020 Winner

Python, Mediapipe, OpenCV

- A gesture-based HCI enabling hands-free mouse and keyboard control using video-based multi-finger tracking and voice input

Vortex 🚩 – Game Engine

Java

- A modular 2D game engine with adherence to OOP principles, display buffering, collision detection, and sprite map parsing

Skills

Languages	Golang, Python, Rust, TypeScript/JavaScript, C/C++, Java, Kotlin, Racket, Solidity, ARM & MIPS Assembly, SQL
Frontend	React, React Native, Tailwind, HTML/CSS, Swing, JavaFX, Figma, Photoshop, Illustrator
Backend	Node, Docker, Kubernetes, AWS, Hardhat, fp-ts, MongoDB, PostgreSQL, MySQL, Flask, FastAPI, Django, Bazel